

world. Relying on Access Modifiers, you can shield both your class's private instance variables and its private methods. With those ends in mind, Java lets review the four different levels of access: private, protected, public and the default if you don't specify - package.

Specifier	Class	Subclass	Package	World
Private	X			
Package	X		X	
Protected	X	X	X	
Public	X	X	X	X

As you can see, a class always has access to its own instance variables, with any access modifier. The second column shows that Subclasses of this class (no matter what package they are in) have access to public (obviously) and protected variables. Therefore, the important point is that Subclasses can reach protected-access variables, but can't reach package-access variables unless the Subclasses happen to be saved in the same package. The third column "package" shows that classes in the same package as the class (regardless of their parentage) have access to data variables. In the fourth column, we see that anything and anyone has access to a public data variable or method—defeating the purpose of encapsulation.

5.2.1 Private

When a class inherits from a Superclass, you cannot access the Superclass's private data variables. The private data is Secret, In other words, the Subclass cannot automatically reach the private data variables of the Superclass. A private data variable is accessible only to the class in which it is defined.

Objects of type Alpha can inspect or modify the iamprivate variable and can call privateMethod, but objects of any other type cannot.

```
class Alpha
{
    private int iamprivate;
    public Alpha( int iam )
    {
        iamprivate = iam;
    }
}
```